NOAA GNSS Radio Occultation Commercial Data Buy
SBEM IDIQ Task Order 1 (TO-1)
Statement of Work

Introduction

In this Task Order, NOAA intends to acquire near-real-time global navigation satellite system (GNSS) radio occultation (RO) satellite data satisfying the requirements set forth in the NOAA Commercial Data Program Space Based Environmental Monitoring (SBEM) IDIQ contract Statement of Work (SOW). NOAA intends to use these data in operational numerical weather prediction models, space weather models, and other systems. NOAA reserves the right to issue subsequent, simultaneous, and additional Task Orders for RO data during this period of performance.

Scope

NOAA is soliciting commercial near-real-time satellite-based GNSS RO and ionospheric measurements that will be processed into neutral atmosphere and space weather products. These derived products will be fed into NOAA's operational data systems, including weather and space weather analysis and prediction systems, and used for weather, climate, and atmospheric research purposes. All data and products will be archived by NOAA. NOAA seeks the following requirements:

Requirements

Part I - Data Preparation

I. Requirements for New GNSS-RO Data Providers

A. For Contractors who have not previously delivered GNSS-RO data to NOAA, the Contractor shall provide all necessary information needed to support NOAA implementation and testing of the Contractor's data. The Contractor shall provide technical documentation (SBEM IDIQ SOW Section C.4.5.3: Data Delivery and Documentation), including but not limited to data format specifications, sample data, and other information required for data ingestion and processing (SBEM IDIQ SOW Sections C.3.1: Test Data, C.3.1.1: Limited Test Data Deliveries, C.4.5.4: GNSS-RO Test Data). Delivery mechanisms and data handling procedures will be communicated to the Contractor.

B. The Contractor shall provide engineering support as needed to enable Government acquisition and processing of the supplied datasets to include supporting technical exchanges. The Contractor shall deliver sample data for secure ingest, configuration and testing. Additionally, the Contractor shall deliver up to 48 hours of contiguous near-real-time data for testing of delivery and data flow processes (SBEM IDIQ SOW

Sections C.3.1: Test Data, C.3.1.1: Limited Test Data Deliveries, C.4.5.4: GNSS-RO Test Data).

C. The Period of Performance for Data Preparation shall be for 35 days from December 1, 2026 through January 5, 2027 and shall start and end at 11:00 am Eastern Time.

Part II - Data Delivery

I. General Requirements for All Data

A. Period of Performance for Data Delivery.

1. For Contractors who have previously provided data to NOAA, the period of performance shall be for 731 days estimated from December 1, 2026 through December 1, 2028 and shall start and end at 11:00 am Eastern Time.
2. For Contractors who have not previously provided data to NOAA, the Data Delivery Period of Performance shall immediately follow the Data Preparation Period of Performance and shall be for 696 days estimated from January 5, 2027 through December 1, 2028 and shall start and end at 11:00 am Eastern Time.

B. The Offeror shall deliver Level 0 and Level 1a data (SBEM IDIQ SOW Section C.4.5: Delivery Requirements).

C. Data delivered shall be subject to the terms and conditions specified in SBEM IDIQ SOW Section C.3.7 : Data Rights . NOAA has no additional obligations under the IDIQ contract or this Delivery Order related to data use by third parties. In no event is NOAA liable to the contractor for third party use or release of contractor data. Delivery of the data will utilize one or more of the following Data Rights options:

1. ***Option 1:*** *Right to distribute all Contractor-provided data to any entity immediately after receipt at NOAA with no restrictions on their use or further distribution, under a Creative Commons Attribution 4.0 International license (CC BY 4.0).*
2. ***Option 2:*** *Right to distribute all Contractor-provided data to U.S. Government agencies, International Partners, including but not limited to those responsible for national meteorological, space weather, or hydrological services, and National Meteorological and Hydrological Services, WMO-designated Regional Specialized Meteorological Centers, members of the Coordination Group for Meteorological Satellites, and non-profit entities and academic entities, immediately after receipt at NOAA, for non-commercial use but not for further distribution.*

3. ***Option 2A:*** *In addition to the rights under Option 2, Option 1 rights will apply to all Contractor-provided data beginning at 24 hours after collection.*
4. ***Option 3:*** *Right to distribute all Contractor-provided data to U.S. Government agencies, International Partners, including but not limited to those responsible for national meteorological, space weather, or hydrological services, and National Meteorological and Hydrological Services, WMO-designated Regional Specialized Meteorological Centers, and members of the Coordination Group for Meteorological Satellites, immediately after receipt at NOAA, for non-commercial use but not for further distribution.*
5. ***Option 3A:*** *In addition to the rights under Option 3, Option 1 rights will apply to all Contractor-provided data beginning at 24 hours after collection.*

D. Spatial distribution is an important aspect of the data delivery requirements. The Government seeks delivery of all global data (all latitudes and longitudes) to meet or exceed the requirements specified in SBEM IDIQ SOW Section C.4.3.2: Specific Geospatial Distributions, paragraphs 1 and 2 only (paragraph 3 does not apply to this task order).

E. This SOW includes Desired Capabilities. Desired Capabilities are features of Offeror provided data that are requested by the Government but are not mandatory. Desired Capabilities are requested in order to improve the utility of delivered data and are considered to add value to the Offeror proposals.

II. Requirement - Neutral Atmosphere Data

A. The Government seeks a total of 10,000 to 26 ,000 compliant radio occultation (RO) profiles per day as specified in SBEM IDIQ SOW Sections C.4.3.3: Neutral Atmosphere Data Requirements (Category 1), and C.4.5: Delivery Requirements.

III. Desired Capability - Total Electron Content (TEC) Data

A. The Government seeks a total of 500 to 1,000 compliant Total Electron Content (TEC) ionospheric tracks per day as specified in SBEM IDIQ SOW Sections C.4.3.5 : Space Weather Data Requirements, C.4.3.5.1 : Requirements for Total Electron Content (TEC), and C.4.5: Delivery Requirements.

IV. Desired Capability - Temporal Resolution Enhancements

A. To increase temporal resolution of the Government's GNSS-RO data collections, the

Government seeks RO observations from Sun-Synchronous Orbits (SSO) other than only within the 0800-1200 and 2000-2400 equatorial crossing time ranges and/or from lower-inclination orbits (non-SSO). The equatorial crossing time is defined as when the space vehicle crosses the Earth's equatorial plane. This Temporal Resolution Enhancement data will be included as part of the total number of profiles in Part II, Section II of this SOW. The Contractor is requested to provide the number of compliant temporal resolution enhancement RO profiles they can deliver within each proposed local time window and/or from each lower-inclined orbit (for satellites not in SSO).

B. While currently a desired capability, NOAA views the diversification of equatorial orbital crossings and increased temporal sampling over the tropics and subtropics as a high-priority strategic objective. The Government intends to use this data to validate requirements for anticipated mandatory coverage in future Task Orders.

V. Requirement Deliverables - Operational Status Reporting

A. The Contractor shall deliver information reports as specified in the SBEM IDIQ SOW Section C.3.3 : Notifications. . The Contractor shall deliver the i nformation reports to NOAA no later than seven (7) calendar days following the conclusion of each three-month reporting period (e.g., the December, January, and February report is due by March 7th; the March, April, May report is due by June 7th; etc.).

B. The Contractor shall inform the Government within 24 hours of any anomalies that adversely affect contractual data delivery requirements and provide an estimated time of repair.